COMP3162-LAB02-620043505

2025-02-03

Lab 2

Question 1

Mean, mode and median values for:

a) Sepal.Length

```
mean(iris$Sepal.Length)
```

```
## [1] 5.843333
```

```
mode_cal(iris$Sepal.Length)
```

```
## [1] 5
```

```
median(iris$Sepal.Length)
```

```
## [1] 5.8
```

b) Sepal.Width

```
mean(iris$Sepal.Width)
```

```
## [1] 3.057333
```

```
mode_cal(iris$Sepal.Width)
```

```
## [1] 3
```

```
median(iris$Sepal.Width)
```

```
## [1] 3
```

c) Petal.Length

```r
mean(iris$Petal.Length)
```

```
## [1] 3.758
```

```r
mode_cal(iris$Petal.Length)
```

```
## [1] 1.4
```

```r
median(iris$Petal.Length)
```

```
## [1] 4.35
```

d) Petal.Width

```r
mean(iris$Petal.Width)
```

```
## [1] 1.199333
```

```r
mode_cal(iris$Petal.Width)
```

```
## [1] 0.2
```

```r
median(iris$Petal.Width)
```

```
## [1] 1.3
```

Question 2

What is the minimum.

a) Sepal.Length

```r
min(iris$Sepal.Length)
```

```
## [1] 4.3
```

b) Sepal.Width

```r
min(iris$Sepal.Width)
```

```
## [1] 2
```

Question 3

What is the largest:

a) Petal.Length

```r
max(iris$Petal.Length)
```

```
## [1] 6.9
```

a) Petal.Width

```r
max(iris$Petal.Width)
```

```
## [1] 2.5
```

Question 4

Using the data in the column named Species, determine:

a) How many species exist in the dataset? (hint: `summary()` and `length()` functions can be used)

Species Exisiting:

```r
summary(iris$Species)
```

```
##     setosa versicolor  virginica 
##         50         50         50
```

Total Count:

```r
length(iris$Species)
```

```
## [1] 150
```

Question 5

Analysis of the Verisicolor Species:

a) Extract all data for the 'versicolor' species and place it in a new object called v.species. *(Hint: use a conditional statement between the square brackets to select the rows that have the matching Species value. Leave the column value empty. An example is in the lecture notes)*

```r
v.species <- iris[iris$Species == "versicolor",]
head(v.species)
```

```
##    Sepal.Length Sepal.Width Petal.Length Petal.Width    Species
## 51          7.0         3.2          4.7         1.4 versicolor
## 52          6.4         3.2          4.5         1.5 versicolor
## 53          6.9         3.1          4.9         1.5 versicolor
## 54          5.5         2.3          4.0         1.3 versicolor
## 55          6.5         2.8          4.6         1.5 versicolor
## 56          5.7         2.8          4.5         1.3 versicolor
```

b) How many of the v.species have both a Sepal.Length and Petal.Length greater than 4? *(you will need to use the hint from part 5a to help you here in addition, consider the nrow() and ncol() functions).*

```r
nrow(v.species[v.species$Sepal.Length > 4 & v.species$Petal.Length > 4,])
```

```
## [1] 34
```

Question 6

Several variations exist between sepals and petals:

a) Add a new column called **avg_dist** to the **iris** object which has the *average distance* for each row as calculated *(Ensure you show the result in your rmarkdown file)*. The *average distance* for a row is calculated as:

`((Sepal.Width - Petal.Width) + (Sepal.Length - Petal.Length))/2`

```r
iris$avg_dist <- ((iris$Sepal.Width - iris$Petal.Width) + (iris$Sepal.Length - iris$Petal.Length))/2
```

b) What is the average distance between the sepal and petals in this species (this is the average of all rows).

```r
mean(iris$avg_dist)
```

```
## [1] 1.971667
```

Question 7

For each species, what is the mean *average distance*? *(requires calculating for each species separately)*

```r
aggregate(iris$avg_dist, by = list(iris$Species), FUN = mean)
```

```
##      Group.1     x
## 1     setosa 3.363
## 2 versicolor 1.560
## 3  virginica 0.992
```

Question 8

Add another column to the dataset called **s.group** which sets each value to either: 0 or 1 based on whether the value in the **avg_dist** column. If the value of **avg_dist** is less than or equal to the *mean average distance*, then the current row in **s.group** should be set it to 0, otherwise set it to 1. *(hint: you can use the ifelse function)*

```r
iris$s.group <- ifelse(iris$avg_dist <= mean(iris$avg_dist), 0, 1)
head(iris)
```

```
##   Sepal.Length Sepal.Width Petal.Length Petal.Width Species avg_dist s.group
## 1          5.1         3.5          1.4         0.2  setosa     3.50       1
## 2          4.9         3.0          1.4         0.2  setosa     3.15       1
## 3          4.7         3.2          1.3         0.2  setosa     3.20       1
## 4          4.6         3.1          1.5         0.2  setosa     3.00       1
## 5          5.0         3.6          1.4         0.2  setosa     3.50       1
## 6          5.4         3.9          1.7         0.4  setosa     3.60       1
```